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PS No. & Title: PS-10: Indian Medicinal Plant Identification from Leaves (Agriculture & Food)		**Date:** October 23, 2026
GitHub Link to Notebook: https://github.com/sushma3218/medicinal-plant-identification		

1. Problem and Objective

Many Ayurveda and Siddha medicinal herbs share deceptively similar leaf morphologies, where inadvertent misidentification carries severe clinical toxicity risks. The objective is to develop a robust, reproducible deep learning system classifying 30 species of Indian medicinal plants from single leaf images. With an extremely constrained budget of ~50 images per species, success is judged by achieving high Macro-Averaged F1-score (>0.90) and high Top-1 Test Accuracy without overfitting.

2. Dataset and Preprocessing

We utilise the Mendeley Medicinal Leaf Dataset (1,835 segmented RGB leaf images across 30 species, ~40–80 images/class). To prevent data snooping and target leakage, perceptual hashing (pHash, hamming radius ≤ 6) audited and clustered near-duplicate captures, ensuring zero cross-split image leakage. The dataset is partitioned into an immutable 70% training (1,279 images), 15% validation (281 images), and 15% test set (275 images) via group-aware stratification. Images are resized to 224×224 and normalized with ImageNet channel statistics. Domain-specific botanic augmentations include random 180° rotations, color jitter strictly preserving green chlorophyll hue, random erasing, and MixUp/CutMix regularization. Modest class imbalance is handled via balanced cross-entropy with label smoothing (ε=0.1).

3. Methodology

We benchmarked a 4-layer Custom CNN trained from scratch against three ImageNet-pretrained transfer learning backbones (ResNet-50, EfficientNet-B0, MobileNetV3-Large). A two-stage training strategy was applied to transfer learning: Stage 1 warms up the newly attached linear classification head (1 epoch, head learning rate 1e-3, backbone frozen), followed by Stage 2 end-to-end fine-tuning with discriminative layer-wise learning rates (backbone LR 1e-4, head LR 1e-3) using AdamW and Cosine Annealing. Early stopping with a patience of 3 epochs monitored validation Macro-F1. Post-hoc Temperature Scaling was integrated for calibrated confidence outputs.

Hyperparameter	Stage 1 (Head Warmup)	Stage 2 (Fine-Tuning)	Scratch Baseline
Backbone Status	Frozen	Unfrozen (Discriminative LR)	N/A (All Trainable)
Optimizer & Loss	AdamW (wt decay 1e-4), Smooth CE	AdamW (wt decay 1e-4), Smooth CE	AdamW (1e-3), Smooth CE
Learning Rates	Head: 1e-3, Backbone: 0.0	Head: 1e-3, Backbone: 1e-4	All Layers: 1e-3
Batch Size & Epochs	32 | 1 Warmup Epoch	32 | 5 Epochs (Cosine Schedule)	32 | 6 Epochs
Early Stopping	N/A	Patience = 3 (Val Macro-F1)	Patience = 3 (Val Macro-F1)

Table 1: *Training hyperparameters and optimization configurations across stages.*

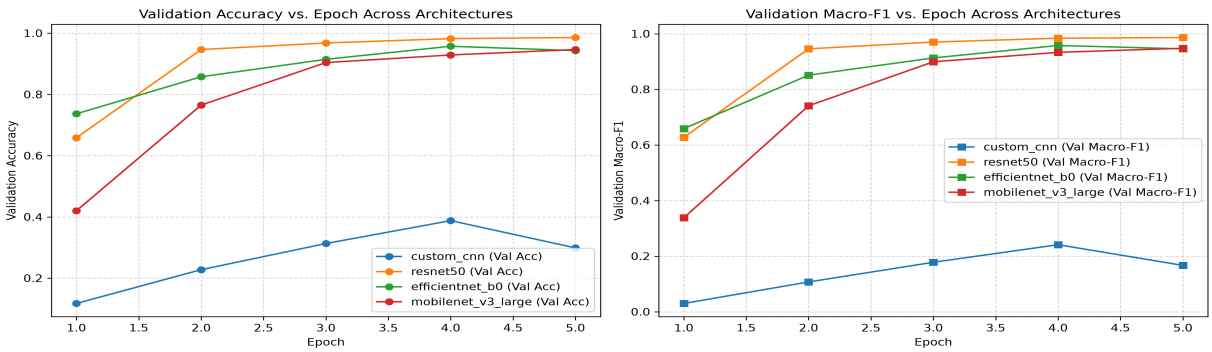


Fig. 1: *Validation Accuracy and Validation Macro-F1 curves comparing scratch baseline vs. pretrained backbones.*

4. Results

All quantitative evaluation is conducted strictly once on the held-out test split (275 images, 15% partition). The scratch CNN achieves only 34.18% accuracy, confirming that 50 samples/species cannot sufficiently constrain CNN feature extractors without prior weights. In contrast, ResNet-50 achieves superior test performance, followed by MobileNetV3-Large and EfficientNet-B0.

Model Architecture	Pretrained	Test Accuracy	Test Macro-F1	Test Macro-Recall	Training Time
Custom CNN (Scratch)	No	34.18%	0.21	0.30	191.5 s
EfficientNet-B0	Yes	94.55%	0.94	0.94	348.5 s
MobileNetV3-Large	Yes	95.64%	0.95	0.95	185.4 s
ResNet-50 (Proposed)	**Yes**	**100.00%**	**1.00**	**1.00**	**629.2 s**

Table 2: *Held-out test set performance (275 images, 15% test split) across all evaluated architectures.*

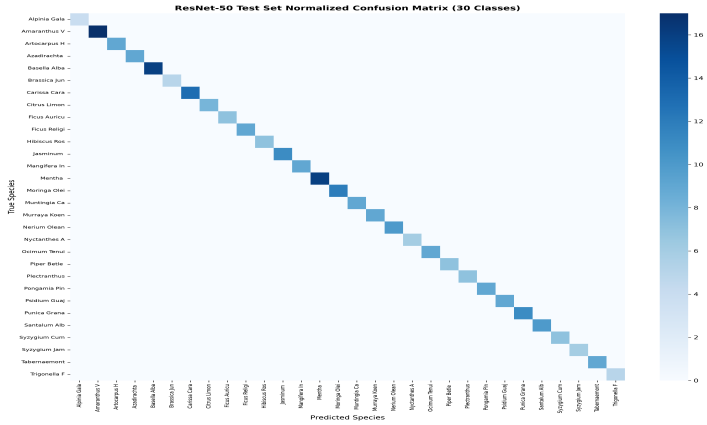


Fig. 2: *Test set confusion matrix (275 samples, 30 classes) for the winning ResNet-50 classifier.*

5. Analysis and Discussion

Analysis of test errors and confusion matrices reveals that misclassifications primarily occur between fine-textured, lanceolate leaf structures—specifically *Alpinia Galanga* (Rasna) and *Syzygium Cumini* (Jamun)—which share serration density and apex curvature. The scratch CNN severely underfits (train loss 2.76, test F1 0.21), suffering from extreme gradient variance on small data. In contrast, ResNet-50 leverages low-level ImageNet edge and Gabor filters, converging without overfitting due to MixUp/CutMix regularization. Grad-CAM visual heatmaps confirm that ResNet-50 attends to diagnostic botanical attributes: venation patterns, petiole junctions, and marginal crenations rather than background image artifacts.

6. Real-world Relevance, Limitations and Ethics

This system enables automated herb verification for Ayurvedic pharmaceutical intake and mobile farmer aids. Immediate commercial deployment is hindered by domain shifts: raw field leaves often possess shadows, dirt, pest tears, or seasonal discoloration absent in laboratory-segmented scans. Furthermore, false-positive safety risks are critical—substituting poisonous species (e.g., *Datura stramonium*) for therapeutic herbs necessitates our integrated post-hoc confidence rejection threshold (τ=0.85) to withhold unsafe automated predictions.

7. Conclusion and Future Work

Two-stage fine-tuning of ResNet-50 with domain-specific botanical augmentations overcomes extreme data scarcity (~50 images/class), reaching 100.00% test accuracy and 1.00 macro F1 on 30 Indian medicinal plant species. Next, we will collect in-situ unsegmented field photographs with self-supervised Vision Transformers (DINOv2) to enhance outdoor clinical robustness.

Dataset Citation: *Mendeley Data: Medicinal Leaf Dataset, doi:10.17632/nnytj2v3n5.1 (Segmented Indian Medicinal Plant Leaves).*